Aditya Kushwaha

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Open to relocation: Bengaluru · Hyderabad · Pune · NCR

SUMMARY

Backend engineer building durable distributed systems in TypeScript, Node.js and PostgreSQL. Seven months at Pazago across two products: **Provue.ai**, a consumer AI platform serving **200K+ users**, and **Mettle**, an autonomous security-review agent where correctness under crash, budget and scope limits is the product. **2,500+ merged commits across 790+ pull requests** in four production repositories since March 2026. B.Tech CS 2026, **CGPA 9.69/10**.

EXPERIENCE

Software Engineer I, Backend · Pazago
Mumbai · Mar 2026 – Present

TypeScript, Node.js, Express, Mastra, PostgreSQL, Prisma, Redis, Bun, Turborepo, AWS

Provue.ai — consumer AI companion platform, 200K+ users

- **Cut AWS spend by ₹15K/month** with per-(persona, user, day) rollups replacing per-request aggregation, covering indexes, denormalised credit counters, removal of a leaderboard sequential scan and staggered cron windows.
- **Own monthly AWS cost attribution for a ~$12K/month multi-tenant account** across nine business units; rebuilt the attribution method on cost-allocation tags and live inventory after Cost and Usage Report delivery failed, reconciling monthly against the invoice.
- **Built the multiplayer games subsystem end to end:** catalog, invite lobbies, matchmaking, entry tiers, scoring, settlement, leaderboards. Made reward claims collision-safe with idempotent bulk inserts; fixed wrong-winner and draw settlement in production.
- **Shipped 20+ LLM agent tools on Mastra** — async music generation (Redis queue and worker with completion notifications), text-to-speech, image generation, a knowledge-base-backed stock verdict tool and PDF report generators — all streamed over SSE.
- **Built credits, billing and usage analytics:** per-turn accounting for messages, tool calls and audio, INR cost per interaction, weekly free credits with a config-driven reset and meter, wallet history, and 15+ retention and payment dashboard endpoints.
- **Hardened group chat and access control:** superseded stale queued turns so personas answer only the sender's latest message; scoped knowledge to the caller; added RBAC across skills.

Mettle — autonomous code-review and penetration-testing agent

- **Designed the durable run layer:** lease-fenced worker ownership with atomic conditional claims, content-addressed dedup keys and an append-only journal, so a SIGKILLed worker resumes exactly where it stopped without duplicating dispatched work.
- **Enforced per-run token, step, wall-clock and USD limits** using integer nano-USD reservation with provider-side reconciliation; unknown cost is recorded as unknown, never as zero.
- **Integrated governed scanners** (Semgrep, Gitleaks, OSV, Nuclei, ZAP, SQLMap, Naabu, Joern) behind tool allow-lists, executing untrusted targets inside gVisor sandboxes with no raw network access beneath the autonomy layer.
- **Built the findings lifecycle:** proposals commit atomically with their tool result, identical replays are idempotent, changed content raises a conflict, and unsettled candidates escalate to human review rather than silently passing.
- **Wrote 100 deterministic evaluation cases** (70 offline, 30 database-backed) plus worker-kill integration tests, a restore drill, and a detection benchmark reporting misses, false positives and confidence intervals.

Backend Developer Intern · Testlify (HNR Tech)
Jun – Aug 2025

- Built REST APIs in Node.js, Express and MongoDB for a skills-assessment platform; added Redis caching on hot read paths; contributed to CI/CD.

EDUCATION

B.Tech, Computer Science · Bharati Vidyapeeth Deemed University, Pune
2022 – 2026 · CGPA 9.69/10

Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning

PROJECTS

Ghidorah — Durable Agent Harness · *TypeScript, Bun, Mastra, PostgreSQL, Zod* github.com/AdityaKushwaha94/Ghidorah

- Open-source proof of concept of the harness above, as a five-package Bun workspace. Mastra supplies the model and tool loop; Ghidorah owns request validation, allow-listed tool execution, budgets, run ownership, an append-only action journal, evidence integrity and ordered frontend events, with product records and native workflow checkpoints in separate PostgreSQL schemas.
- Provider-neutral model gateway over OpenRouter with a pinned live GLM-4.7 route. Verified by a full local pipeline: contract and schema-drift checks, 10 recovery tests, integration tests that kill real workers and assert checkpoint fencing and wall deadlines, a dump-and-restore drill, and **100 deterministic evaluations** (70 offline, 30 database-backed).

LLM Evaluation & Observability Service · *Python, FastAPI, Redis (RQ), PostgreSQL* github.com/AdityaKushwaha94/llm-eval-observability

- Trace ingestion API returning 202 immediately and scoring off the request path: deterministic checks plus an LLM judge run by a worker, reporting trend, worst offender and version-to-version regression.

Oral Squamous Cell Carcinoma Detection · *Python, TensorFlow, VGG16* github.com/AdityaKushwaha94/oscc-detection

- Fine-tuned a VGG16 CNN for histopathology image classification to **96% test accuracy** with a reproducible training pipeline. Provisional patent filed.

SKILLS

Languages	TypeScript, JavaScript, Python, SQL
Backend	Node.js, Express, Prisma, PostgreSQL, MongoDB, Redis, REST, SSE/WebSockets, JWT, Zod, job queues, distributed workers
AI systems	Mastra, LangChain / LangGraph, OpenRouter, LLM agent tooling, evaluation harnesses, TensorFlow
Infra & tools	AWS (EC2, S3, Lambda, SSM, Cost Explorer), Docker, gVisor, GitHub Actions, Bun, Turborepo, Git, Linear, Jira, Linux

ACHIEVEMENTS

- Smart India Hackathon 2023: built EvalMate, an AI-assisted interview platform for DRDO (PS 1653), qualified college round. Tech Expression 2025: top 10 in technical problem solving. Grid 6.0 (Flipkart): shipped a full-stack solution.
- Google Developer Student Club winner 2023. Presented on software and AI standardisation at a Bureau of Indian Standards forum.